IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

Objective: Predict the success of Falcon 9 first-stage landings to determine launch price.

Methodology: Data collection via API and Web Scraping, EDA using SQL and Folium, Predictive modeling using 4 classification algorithms (Logistic Regression, SVM, KNN, Decision trees).

Key Finding: All models achieved high accuracy (~83% to 89%), with the Decision Tree being the top performer.

Introduction

SpaceX cost: $62 Million. Others: $165 Million+.

Savings are driven by **reusability**.

Problem: If we can predict the landing, we can predict the cost.

The goal is to build a predictive model that answers:

- Feature Impact: How do factors like payload mass, orbit, and launch site affect landing success?
- Model Selection: Which algorithm (Logistic Regression, SVM, or Decision Tree) is most accurate?
- Optimization: Can hyperparameter tuning via GridSearchCV improve our prediction reliability?
- Cost Estimation: Can we accurately forecast if a stage will be recovered to determine the final price of a mission?

Section 1

Methodology

Methodology

Data Collection & Wrangling

API: Extracted rocket, payload, and orbit data from SpaceX API.

Scraping: Used BeautifulSoup to pull historical launch records from Wikipedia.

Wrangling: Filtered for Falcon 9, handled missing values (PayloadMass mean), and created the **Class** target variable (1=Success, 0=Fail).

EDA & Interactive Analysis

SQL: Used for querying success rates and landing outcomes.

Folium: Created maps to analyze launch site proximity to coastlines and cities.

Plotly Dash: Built an interactive dashboard to filter results by payload and site.

Predictive Analysis

Preprocessing: Features were standardized using StandardScaler.

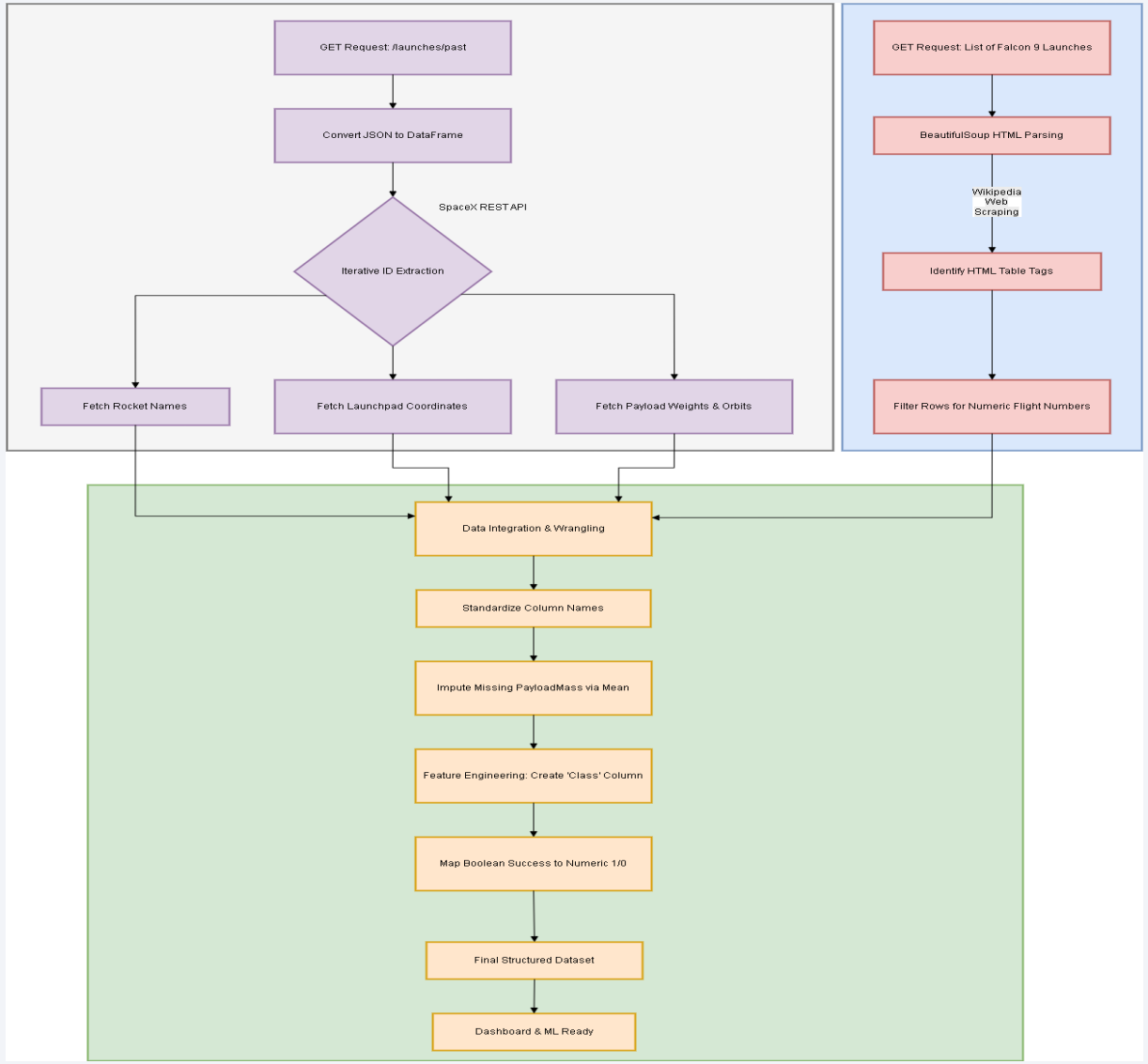
Models: Logistic Regression, SVM, Decision Tree, KNN.

Optimization: GridSearchCV with cv=10.

Data Collection

The data collection process was conducted by:

- Collected launch data from two sources: SpaceX API and Wikipedia launch tables.
- API path captured rocket, payload, orbit, launch site, and core-related fields.
- Web scraping path targeted historical Falcon 9 / Falcon Heavy launch records.
- Combined collection outputs were prepared for downstream wrangling.



Data Collection – SpaceX API

1. Sent GET requests to the SpaceX API / static JSON source.

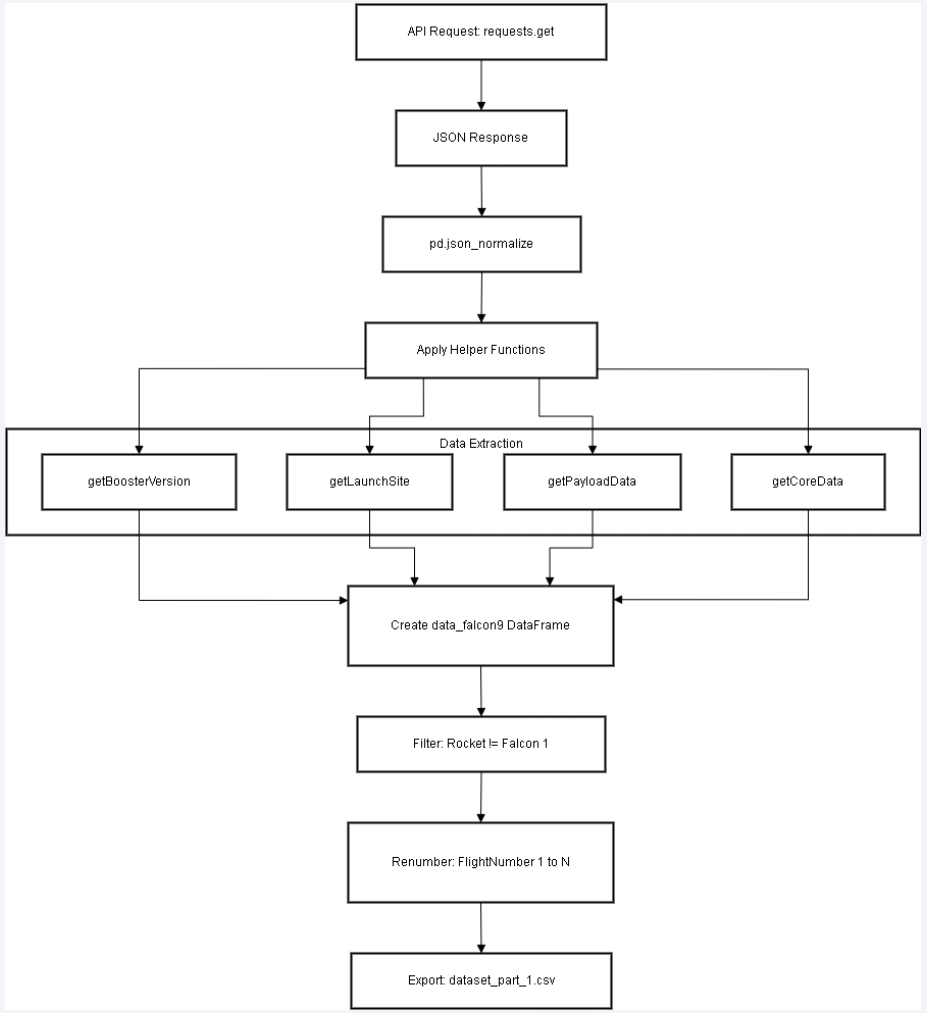
2. Normalized JSON into a tabular dataframe.

3. Extracted booster version, launch site, payload mass, orbit, and core details through helper functions.

4. Filtered to single-core, single-payload launches before narrowing to Falcon 9 missions.

5. Saved the cleaned API-derived dataset as dataset_part_1.csv.

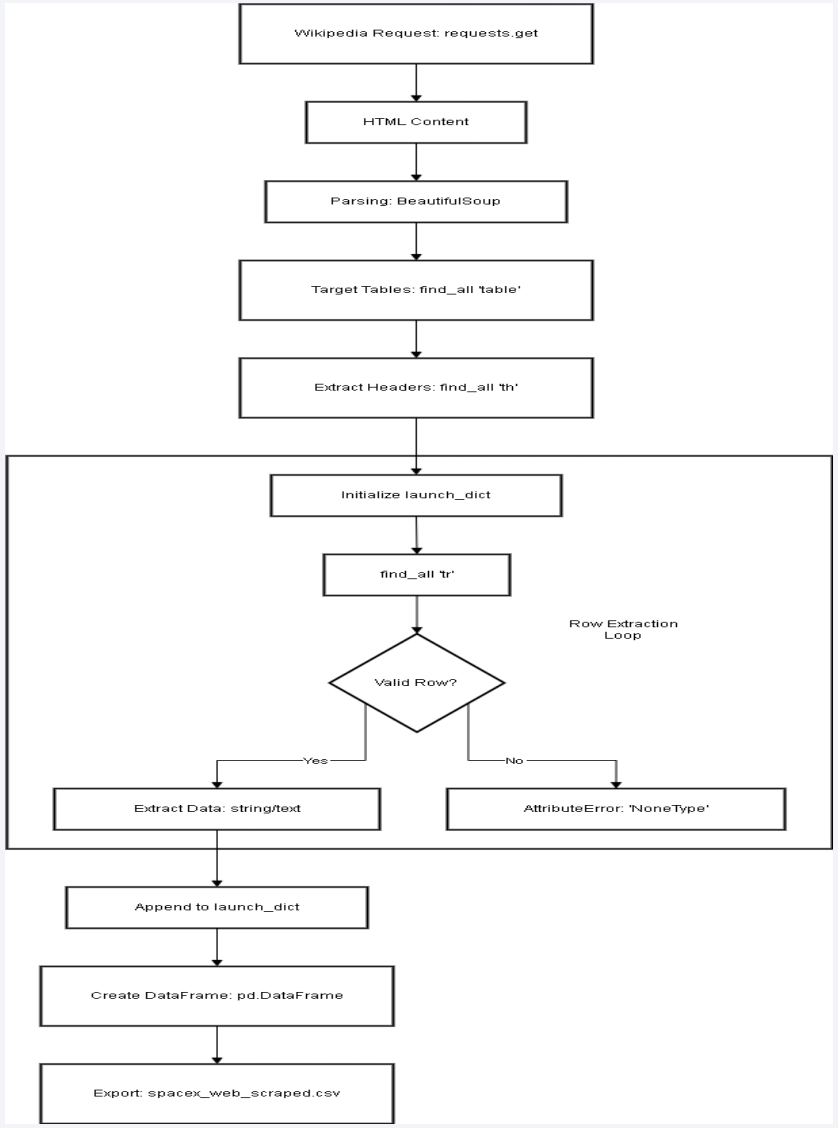
GitHub URL: https://github.com/FadulSaif/IBM-Applied-DataScience-Course.git



Data Collection - Scraping

- Requested the Wikipedia Falcon launch page using requests.
- Parsed the HTML with BeautifulSoup.
- Collected table headers using find_all.Initialized launch_dict and attempted row-by-row extraction from launch tables.
- TODO: Final parsed dataframe output is not available because the notebook currently errors during row extraction.

GitHub URL: https://github.com/FadulSaif/IBM-Applied-DataScience-Course.git



Data Wrangling

- Loaded the cleaned launch dataset into pandas.
- Checked null percentages and data types.
- Reviewed launch-site and orbit distributions.
- Created a target column Class intended to encode landing success.
- **Engineered the Target Variable (Class):**
- Analyzed the Outcome column to distinguish between successful (e.g., True ASDS) and unsuccessful (e.g., False ASDS, None None) landings.
- Mapped successful outcomes to **1** and all others (failures or no attempts) to **0** using a categorical mapping loop.
- Verified the binary distribution to ensure it accurately reflects the SpaceX historical landing success rate (approx. **67%**).

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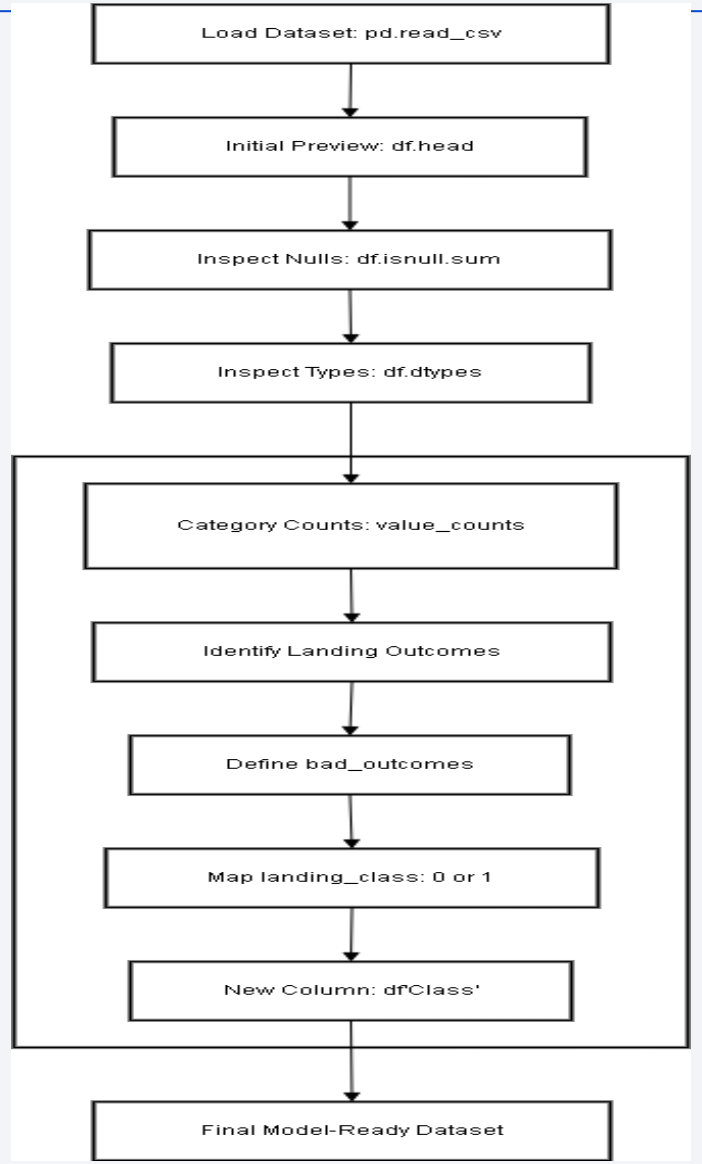
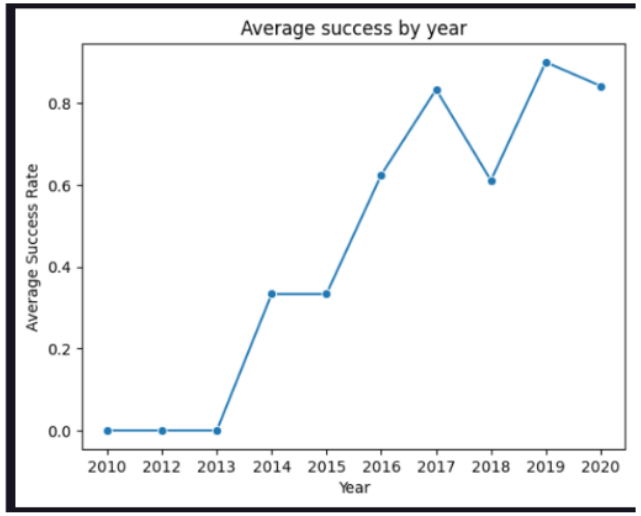
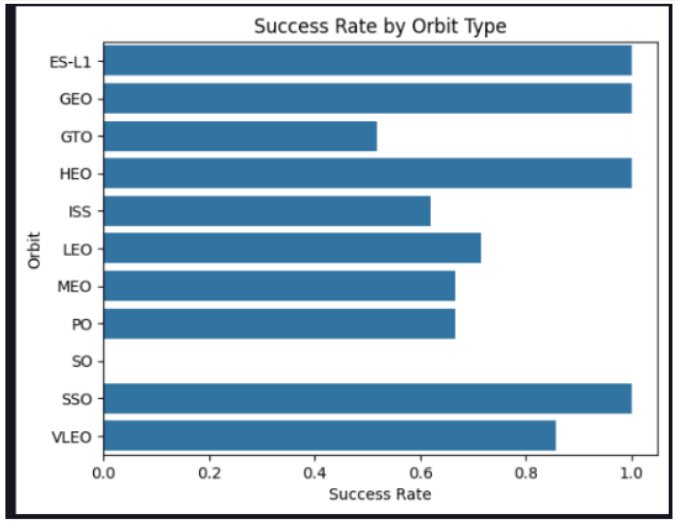


EDA with Data Visualization

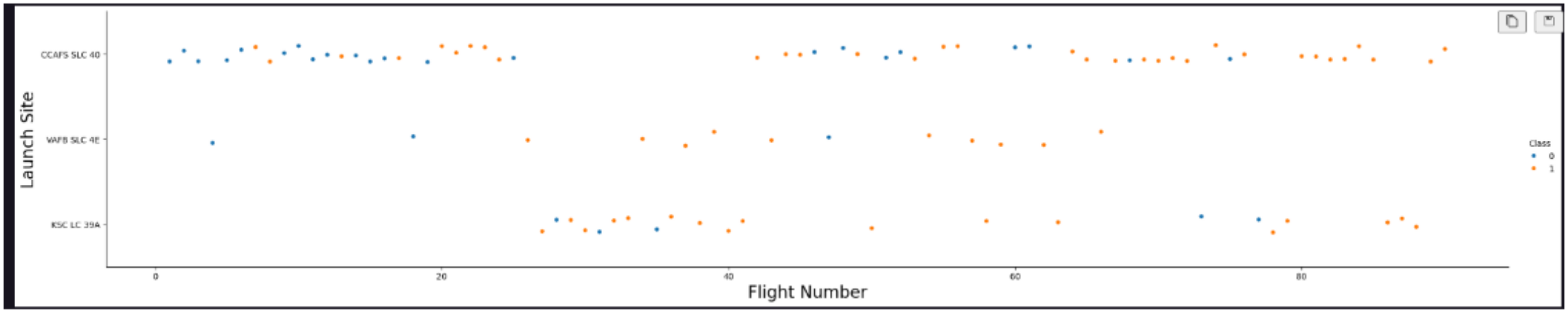


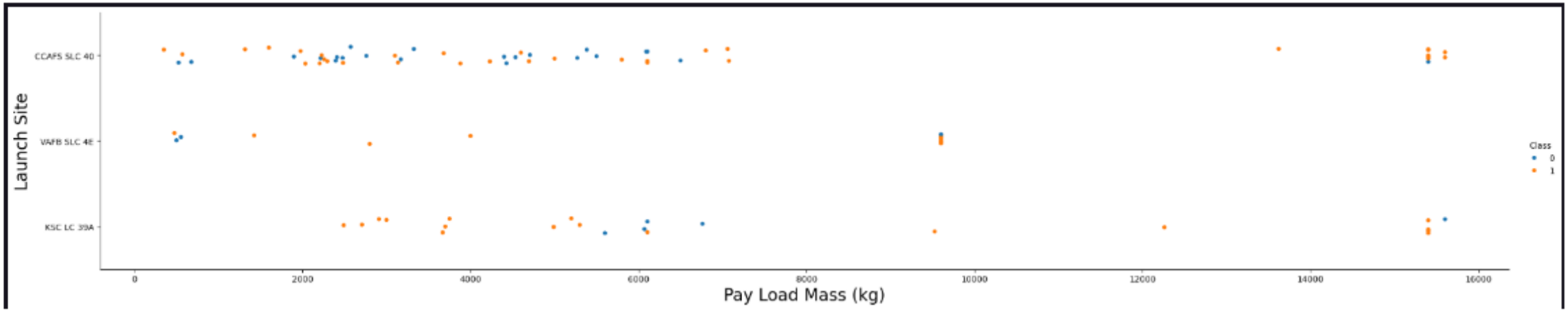


- Plotted scatter charts to study relationships between **Flight Number**, **Payload Mass**, **Launch Site**, and **Orbit** against landing outcome.
- Used a **bar chart** to compare success rates across orbit types.
- Used a **line chart** to examine yearly average landing success trend.
- These charts were used to identify patterns before feature engineering and modeling.

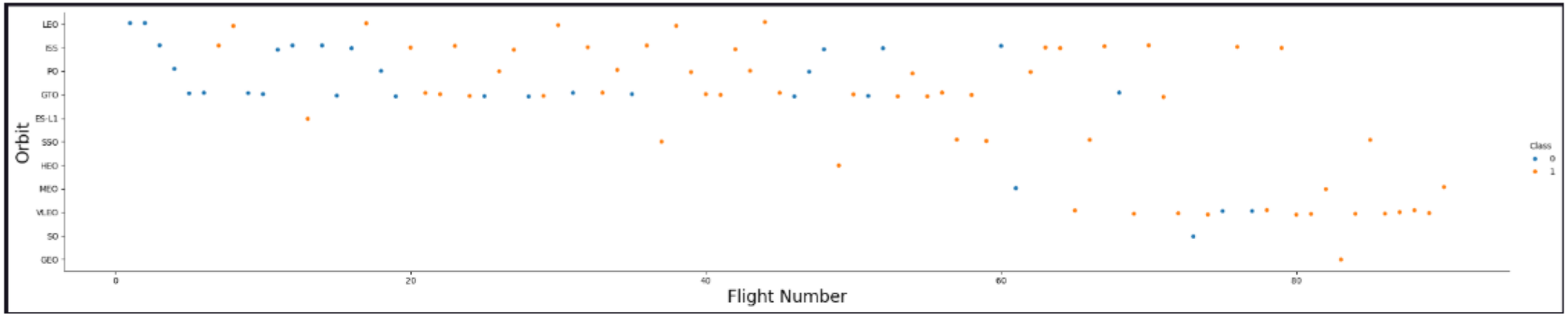
- GitHub URL: https://github.com/FadulSaif/IBM-Applied-DataScience-Course.git

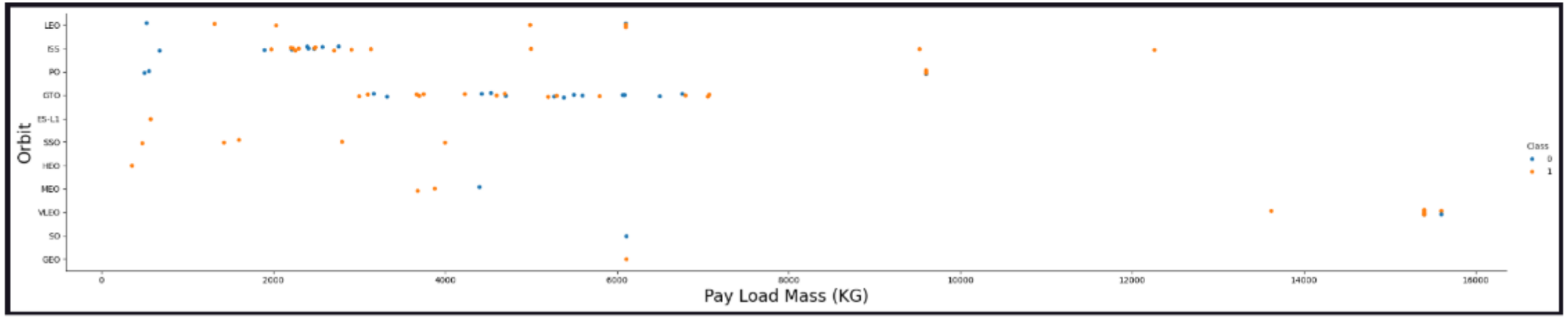
EDA with Data Visualization





EDA with Data Visualization





EDA with SQL

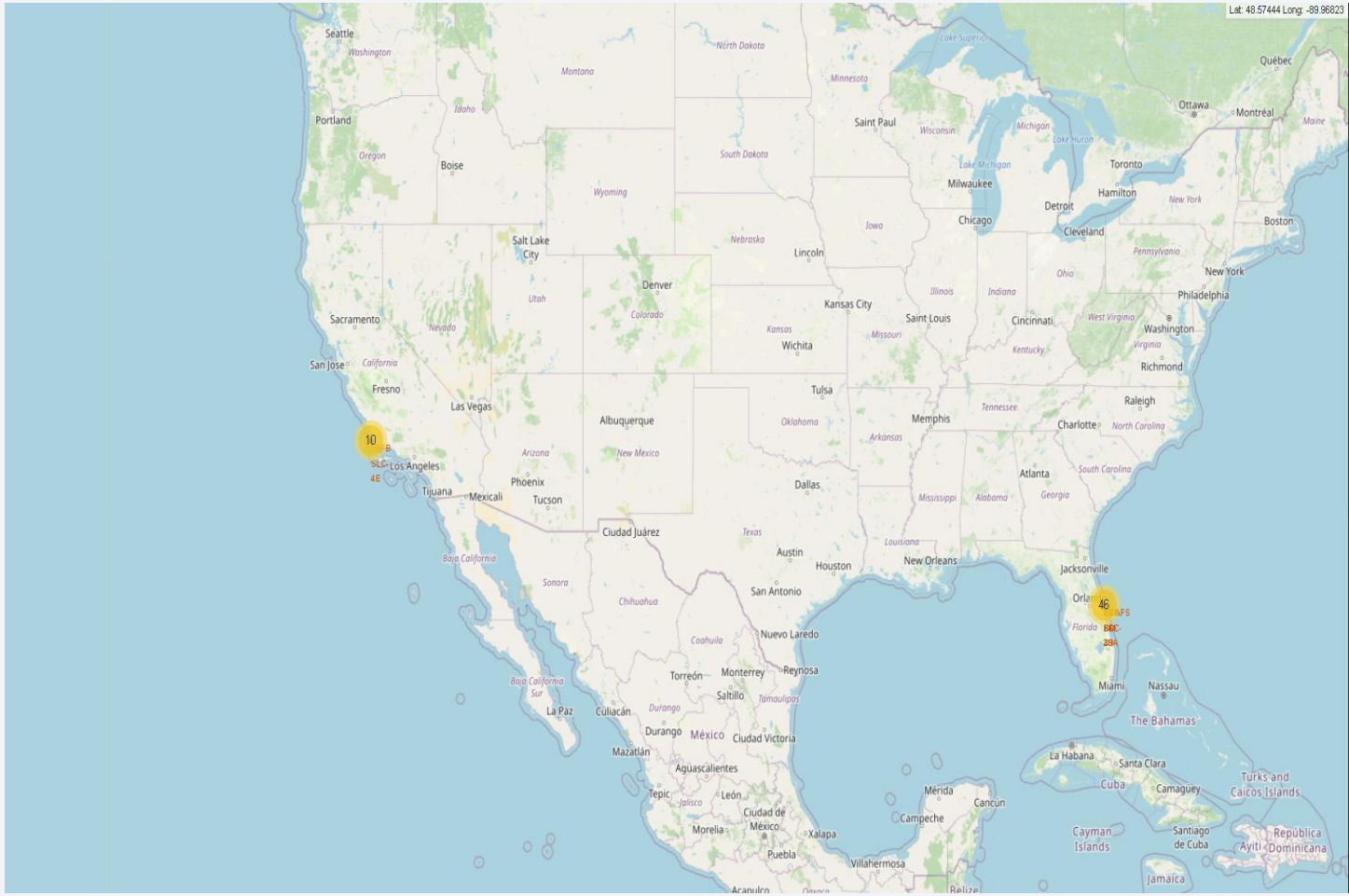
- Queried **distinct launch sites**.
- Filtered launch sites using string matching for **CCA**.
- Computed **total payload mass** for NASA (CRS) missions.
- Computed **average payload mass** for booster version **F9 v1.1**.
- Retrieved the **first successful mission date**.
- Listed boosters for **successful drone ship landings** with payload between **4000 and 6000 kg**.
- Counted mission outcomes, maximum-payload boosters, failed 2015 drone-ship records, and ranked outcomes in a date range.

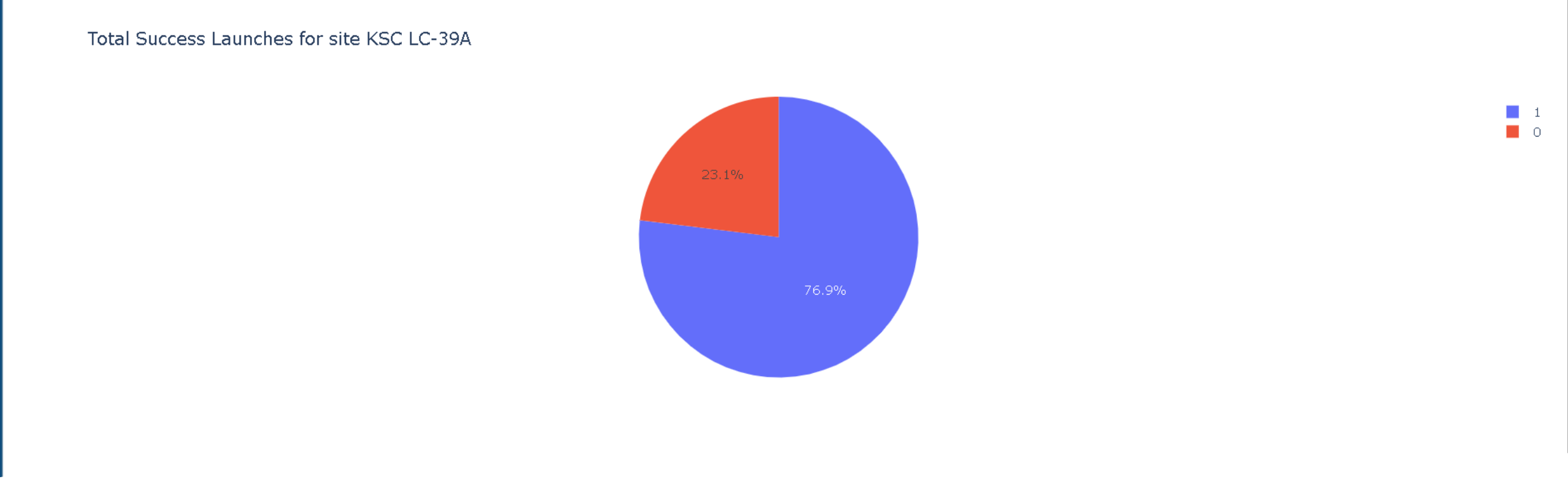
GitHub URL: https://github.com/FadulSaif/IBM-Applied-DataScience-Course.git

Build an Interactive Map with Folium

- Added Circle markers and text labels to show launch-site locations.
- Added MarkerCluster markers to visualize launch outcomes at the same coordinates.
- Added MousePosition to inspect coordinates interactively.
- Added PolyLine and distance labels to analyze launch-site proximity to coastline and nearby features.
- These objects help compare site locations and explore geographic factors related to launches.

GitHub URL: https://github.com/FadulSaif/IBM-Applied-DataScience-Course.git





Build a Dashboard with Plotly Dash

- Built a Dash dashboard with a launch-site dropdown and payload range slider.
- Added a pie chart to show total successful launches for all sites or success-vs-failure for a selected site.
- Added a scatter plot of Payload Mass (kg) vs class, colored by Booster Version Category.
- These interactions support quick filtering by site and payload range.
- GitHub URL: https://github.com/FadulSaif/IBM-Applied-DataScience-Course.git

Predictive Analysis (Classification)

- Loaded engineered features and converted Class to target variable Y.
- Standardized features using StandardScaler.
- Split the dataset into training and testing sets with test_size = 0.2 and random_state = 2.
- Trained and tuned Logistic Regression, SVM, Decision Tree, and KNN using GridSearchCV (cv = 10).
- Compared models using both validation and test accuracy, then selected the top test-accuracy model set.

GitHub URL: https://github.com/FadulSaif/IBM-Applied-DataScience-Course.git

```
tuned hpyerparameters :(best parameters)  {'C': 1.0, 'gamma': 0.03162277660168379, 'kernel': 'sigmoid'}
SVM accuracy : 0.8482142857142856
```

```
tuned hpyerparameters :(best parameters)  {'C': 0.01, 'penalty': 'l2', 'solver': 'lbfgs'}
Logistic Regression accuracy : 0.8464285714285713
```

```
tuned hpyerparameters :(best parameters)  {'algorithm': 'auto', 'n_neighbors': 10, 'p': 1}
KNN accuracy : 0.8482142857142858
```

```
tuned hpyerparameters :(best parameters)  {'criterion': 'gini', 'max_depth': 6, 'max_features': 'sqrt', 'min_samples_leaf': 1, 'min_samples_split': 5, 'splitter': 'best'}
Decision Tree  accuracy : 0.8892857142857145
```

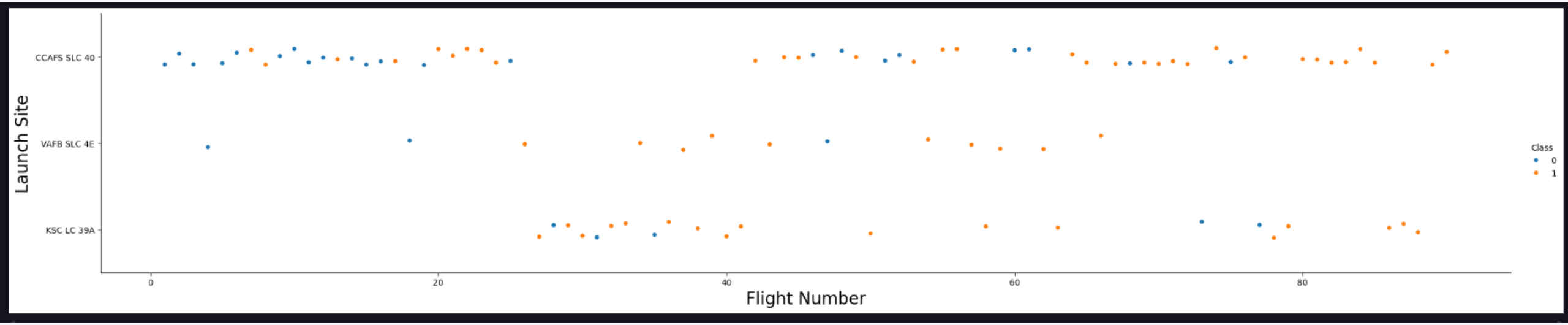
Results

- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

Section 2

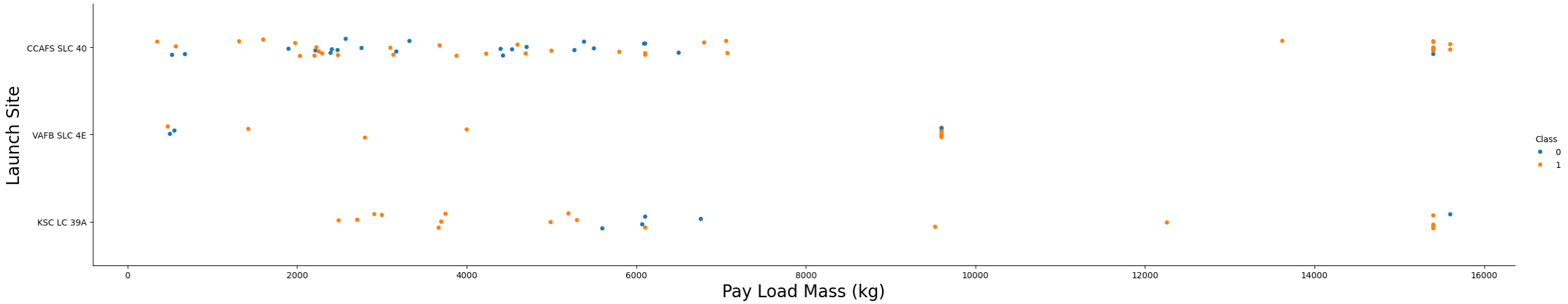
Insights drawn from EDA

Flight Number vs. Launch Site



- Scatter plot of Flight Number vs Launch Site, with color showing landing outcome.
- Used to see whether later missions at each site show more successful landings.
- The notebook states that, overall, as flight number increases, first-stage landing becomes more likely to succeed.

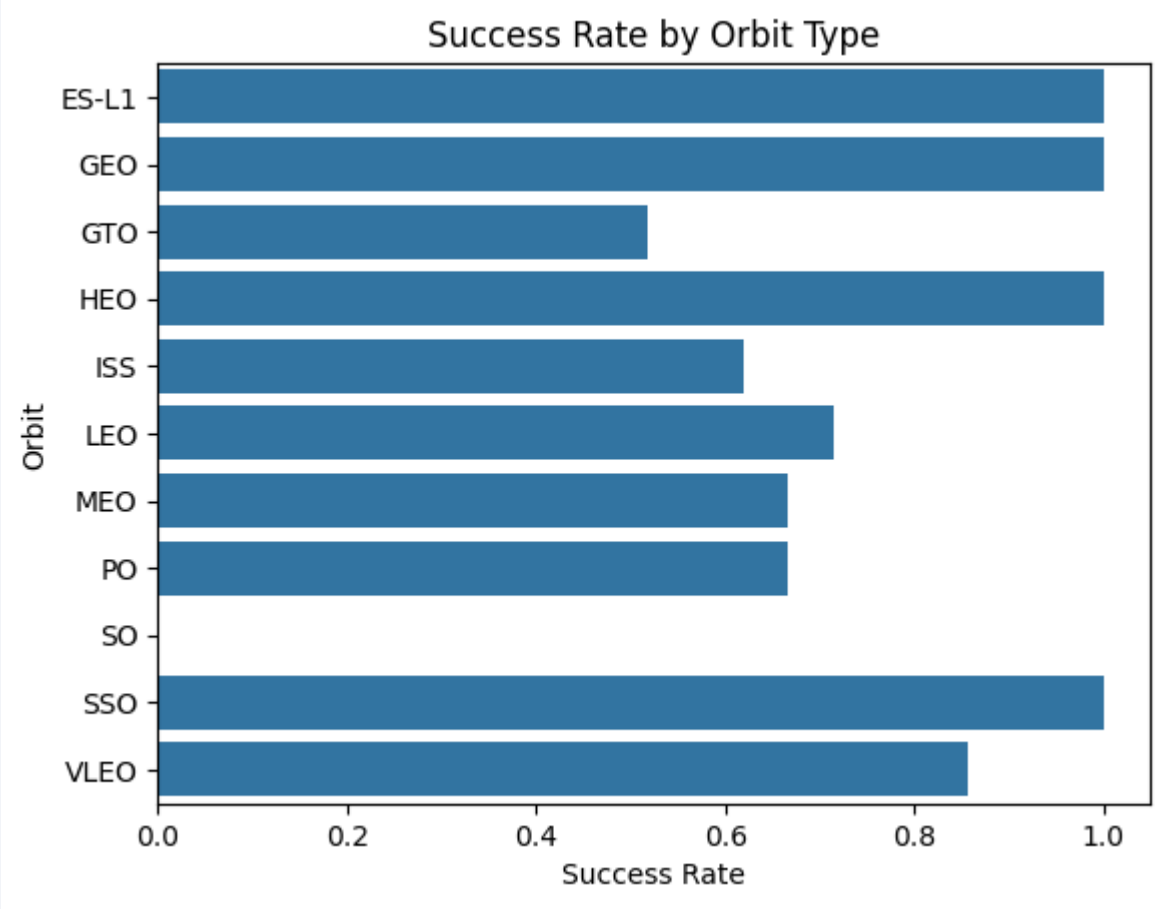
Payload vs. Launch Site



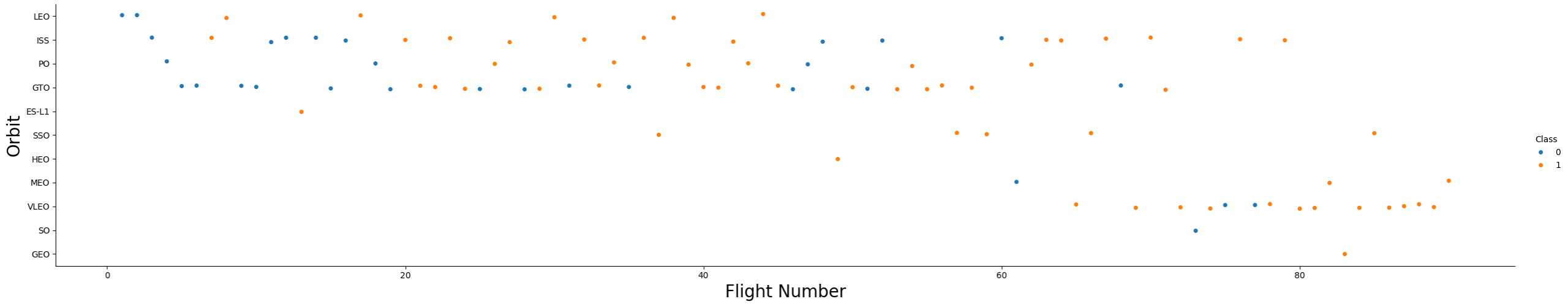
- Scatter plot of **Payload Mass vs Launch Site**, with color showing landing outcome.
- Used to check whether some launch sites handle heavier payload missions.
- The notebook notes that **VAFB SLC** does not show launches with heavy payload mass greater than **10000 kg**.

Success Rate vs. Orbit Type

- Bar chart of **success rate by orbit type**.
- Used to compare which orbit categories are associated with stronger landing outcomes.
- The chart is intended to visually highlight orbit-level differences in landing success.

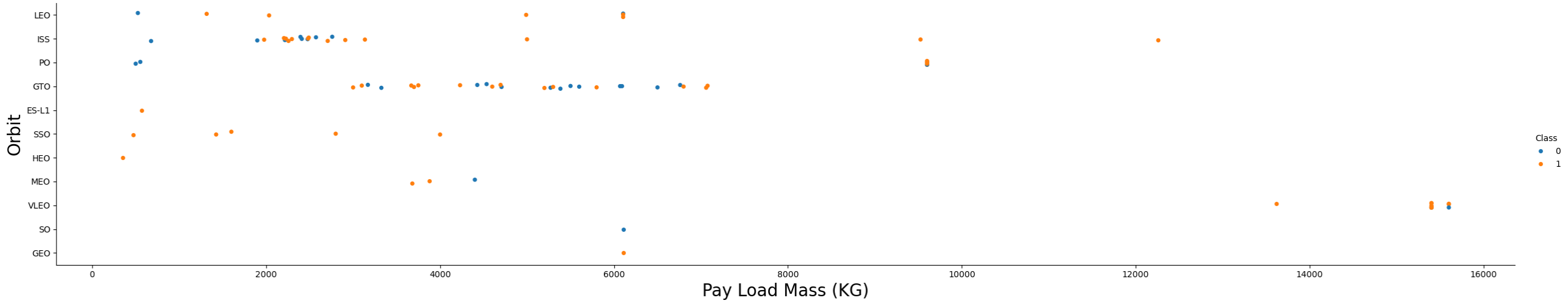


Flight Number vs. Orbit Type



- Scatter plot of Flight Number vs Orbit Type, with color showing landing outcome.
- Used to inspect whether success improves over time within different orbit categories.
- The notebook notes that in LEO, success appears related to number of flights, while in GTO there appears to be no clear relationship.

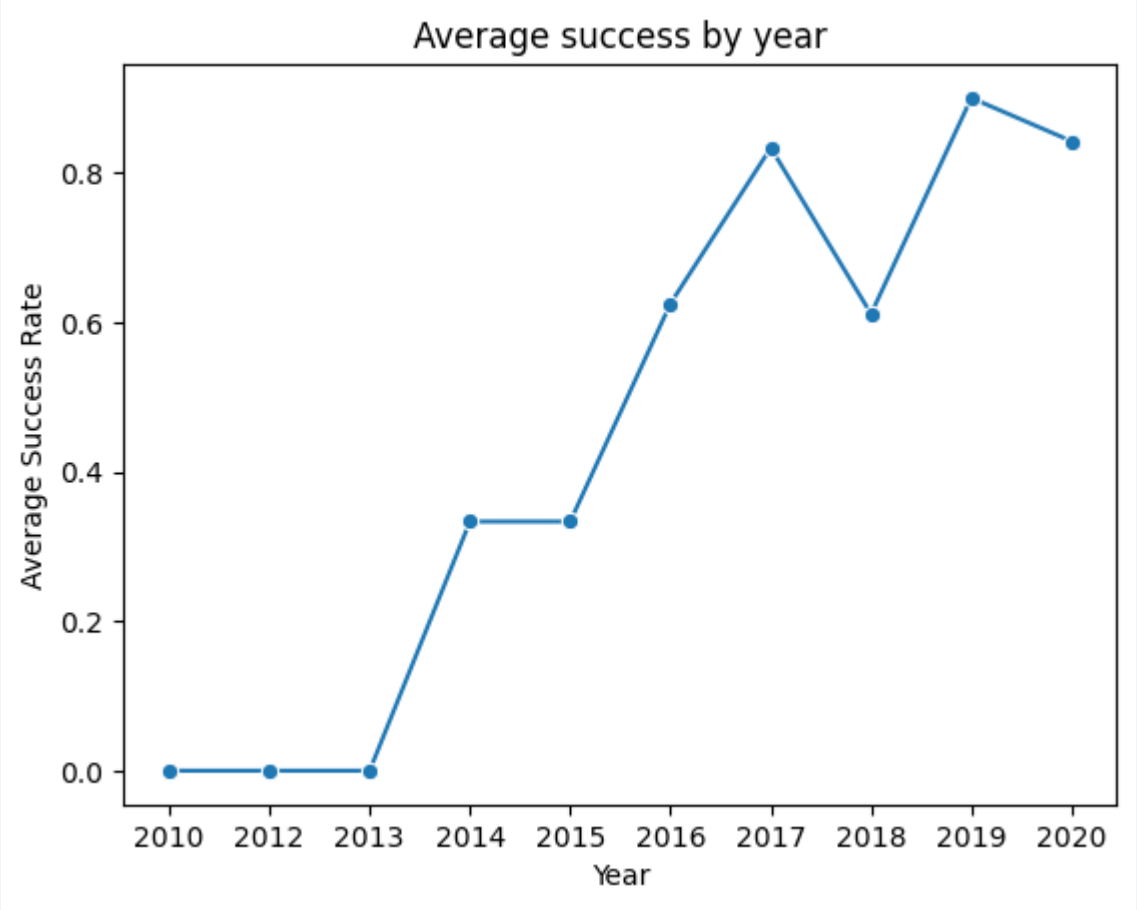
Payload vs. Orbit Type



- Scatter plot of Payload Mass vs Orbit Type, with color showing landing outcome.
- Used to inspect whether heavy payload missions remain successful in different orbit classes.
- The notebook states that with heavy payloads, successful landings are more common for Polar, LEO, and ISS, while GTO contains both successful and unsuccessful outcomes.

Launch Success Yearly Trend

- Line chart of yearly average success rate.
- Used to track whether Falcon 9 landing performance improves over time.
- The notebook explicitly states that the success rate kept increasing from 2013 to 2020.



All Launch Site Names

Unique launch sites returned by SQL:

1. CCAFS LC-40
2. VAFB SLC-4E
3. KSC LC-39A
4. CCAFS SLC-40

```
%sql SELECT DISTINCT LAUNCH_SITE FROM SPACEXTABLE;
```

```
 * sqlite:///my_data1.db
Done.
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Launch Site Names Begin with 'CCA'

- Query filtered launch sites using LIKE '%CCA%'.
- The first five returned records are all **CCAFS LC-40**.
- This confirms repeated CCAFS-based launch entries in the table.

```
%sql SELECT LAUNCH_SITE FROM SPACEXTABLE WHERE LAUNCH_SITE LIKE '%CCA%' LIMIT 5;
```

```
 * sqlite:///my_data1.db
Done.
```

Launch_Site
CCAFS LC-40
CCAFS LC-40
CCAFS LC-40
CCAFS LC-40
CCAFS LC-40

Total Payload Mass

- SQL sum of PAYLOAD_MASS__KG_ for customer NASA (CRS) equals 45,596 kg.
- This query aggregates total carried payload for that customer group.

Total_Payload
45596

Average Payload Mass by F9 v1.1

- SQL average payload mass for F9 v1.1 is 2928.4 kg.
- This summarizes the typical payload associated with that booster version.

Avg_total_payload
2928.4

First Successful Ground Landing Date

- The query result returns 2010-06-04 as the earliest successful mission date.
- This slide shows how SQL was used to retrieve the minimum date satisfying the success condition.

min(Date)
2010-06-04

Successful Drone Ship Landing with Payload between 4000 and 6000

- SQL listed booster versions that successfully landed on a drone ship with payload mass between 4000 and 6000 kg.

Example returned boosters include:

- F9 v1.1
- F9 v1.1 B1011
- F9 v1.1 B1014
- F9 v1.1 B1016
- F9 FT B1020

The full result contains additional FT, B4, and B5 booster entries.

Booster_Version
F9 v1.1
F9 v1.1 B1011
F9 v1.1 B1014
F9 v1.1 B1016
F9 FT B1020
F9 FT B1022
F9 FT B1026
F9 FT B1030
F9 FT B1021.2
F9 FT B1032.1
F9 B4 B1040.1
F9 FT B1031.2
F9 FT B1032.2
F9 B4 B1040.2
F9 B5 B1046.2
F9 B5 B1047.2
F9 B5 B1048.3
F9 B5 B1051.2
F9 B5B1060.1
F9 B5 B1058.2
F9 B5B1062.1

Total Number of Successful and Failure Mission Outcomes

- SQL grouped mission outcomes and counted each category.

Returned counts include:

- Success: 99
- Failure (in flight): 1
- Success (payload status unclear): 1

Success is by far the dominant mission outcome in the table.

Mission_Outcome	Total_number_of_outcomes
Failure (in flight)	1
Success	99
Success (payload status unclear)	1

Boosters Carried Maximum Payload

- SQL returned multiple booster versions tied to the maximum payload mass.

Example returned boosters include:

- F9 B5 B1048.4
- F9 B5 B1049.4
- F9 B5 B1051.3
- F9 B5 B1056.4

This indicates the maximum payload was achieved by more than one booster.

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

2015 Launch Records

SQL returned one matching 2015 record:

- Month: June
- Booster Version: F9 v1.1 B1018
- Mission Outcome: Failure (in flight)
- Launch Site: CCAFS LC-40

Month_Name	Booster_Version	Mission_Outcome	Launch_Site
June	F9 v1.1 B1018	Failure (in flight)	CCAFS LC-40

This slide shows filtered querying by year, outcome, and launch-site details.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- SQL ranked landing outcomes between 2010-06-04 and 2017-03-20.

Returned counts:

- Success: 30
- Failure (in flight): 1

Success is the most common outcome in this time window.

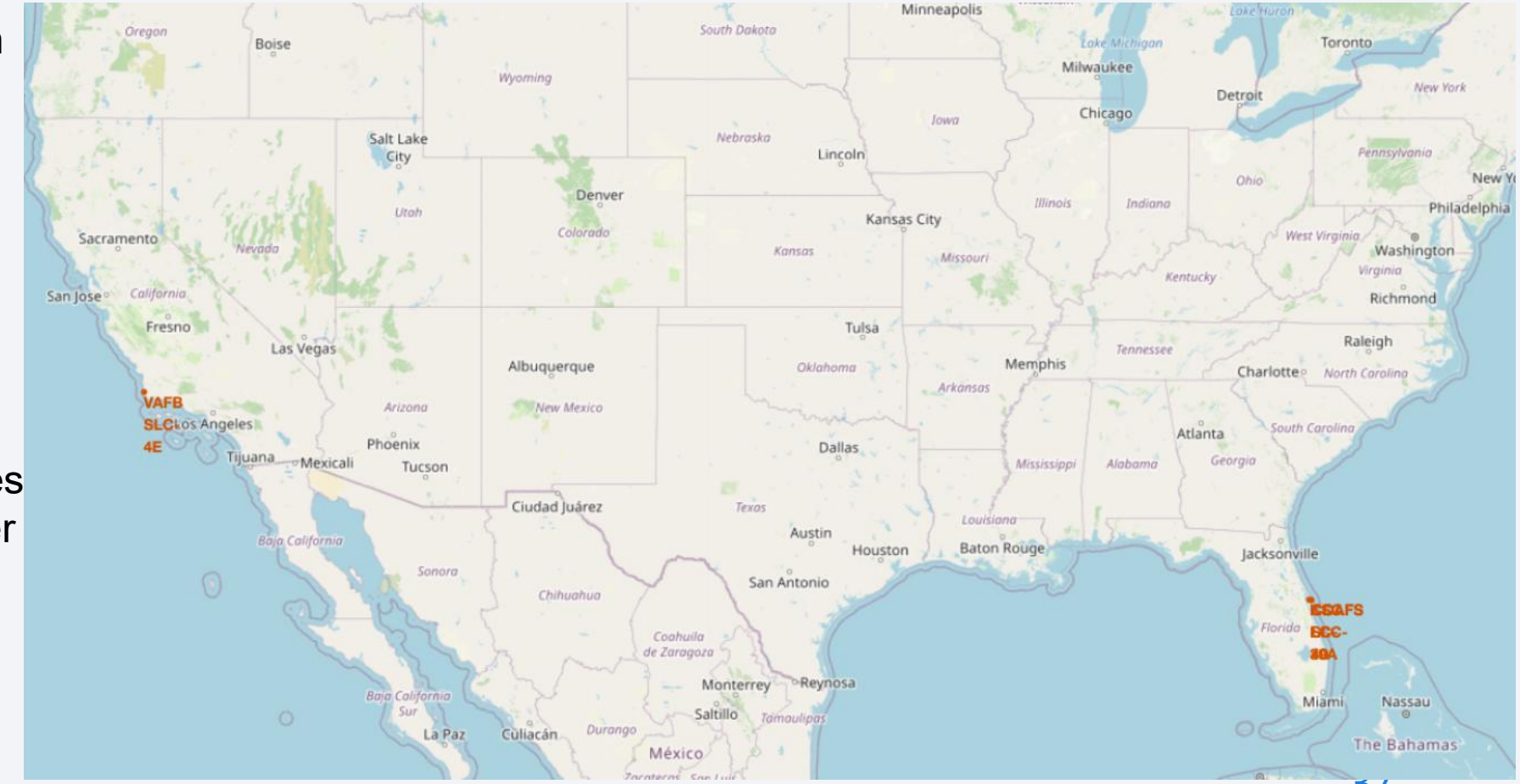
Mission_Outcome	count(*)
Success	30
Failure (in flight)	1

Section 3

Launch Sites Proximities Analysis

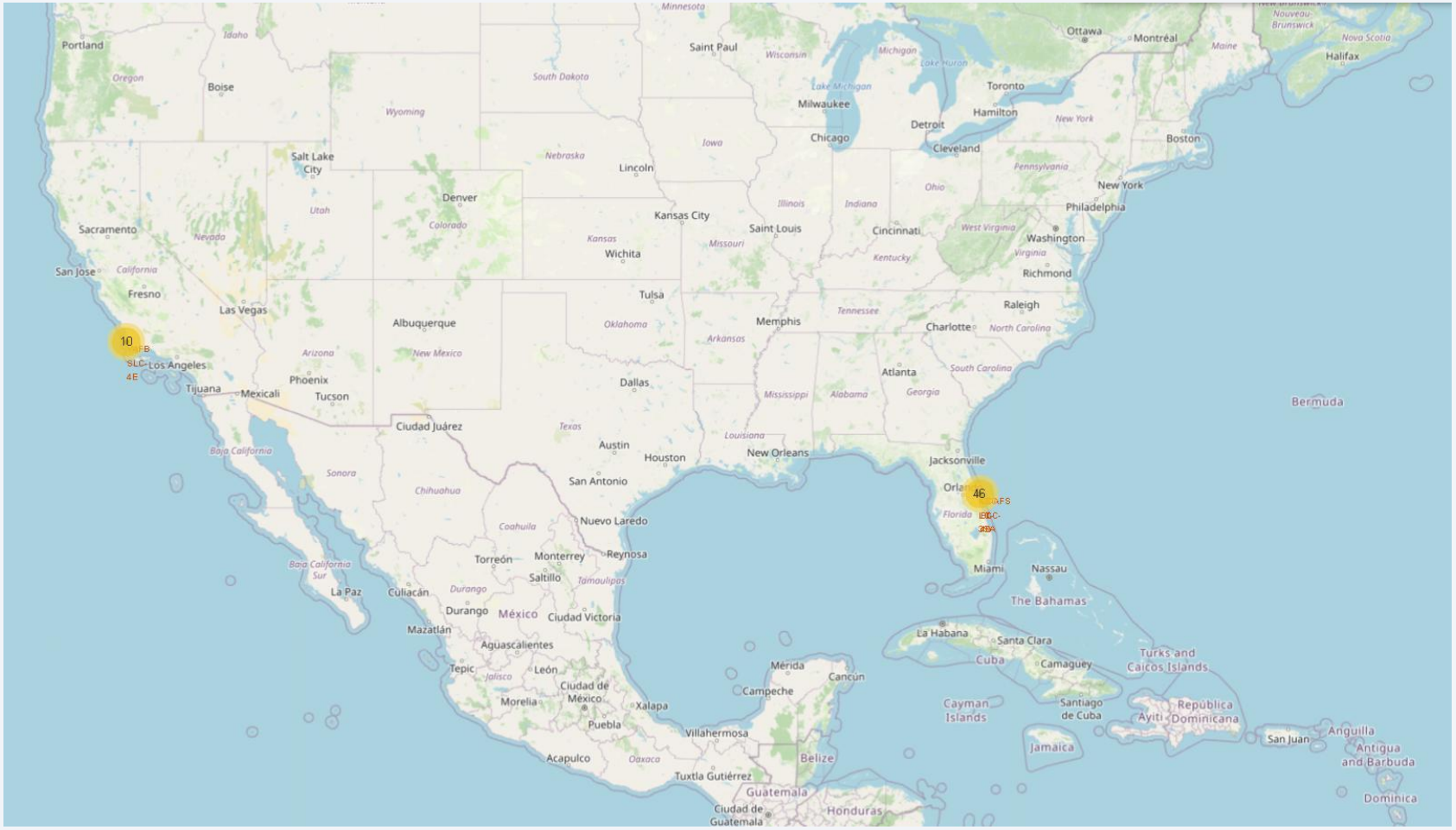
Global Distribution of SpaceX Launch Sites

- The map marks all launch sites using **Circle** objects and text labels.
- It shows that the project uses geographic coordinates to visualize launch-site distribution.
- This screenshot introduces the spatial context for later proximity analysis.



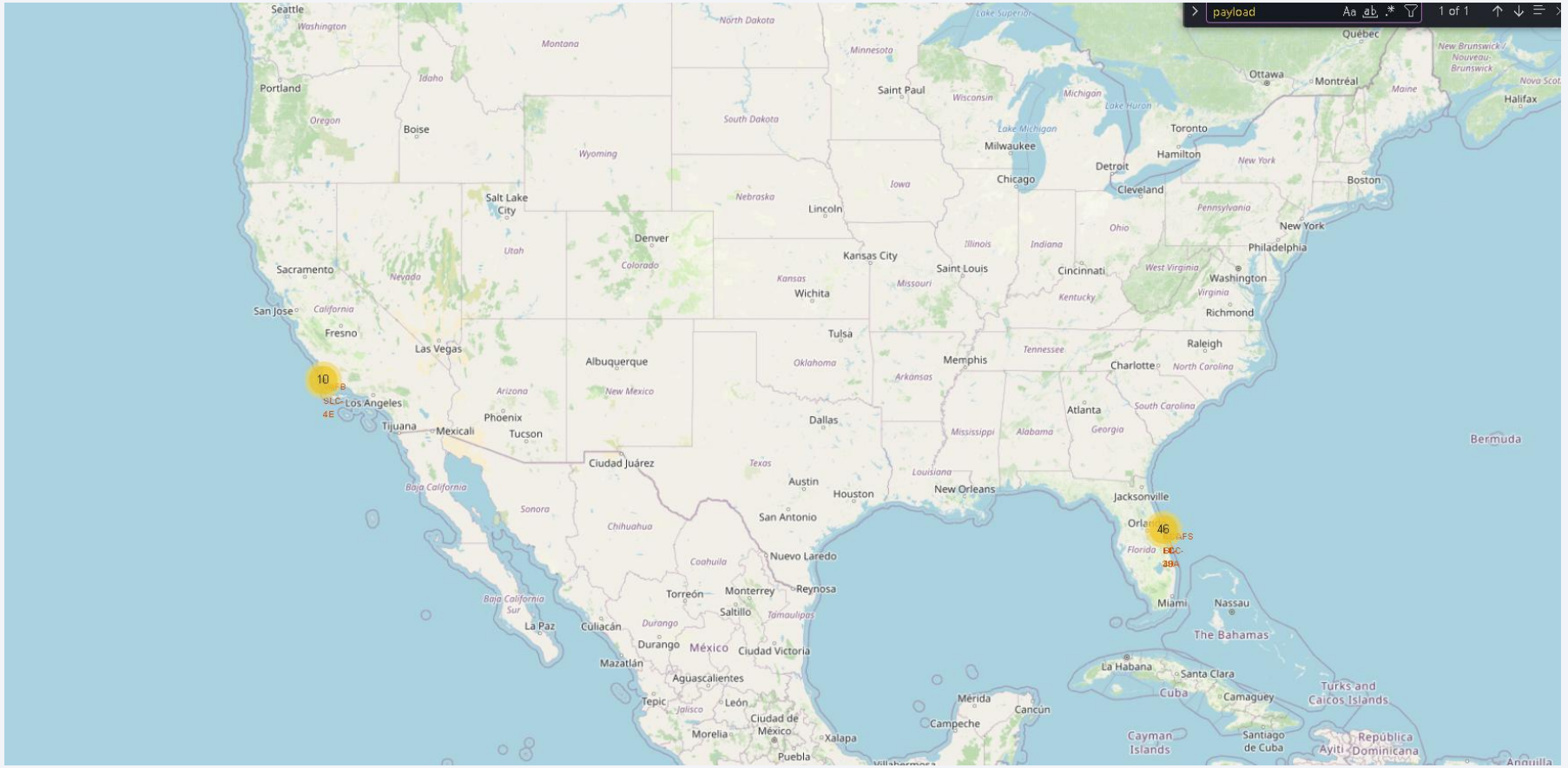
Launch Outcomes by Site

- Marker colors are assigned using the class column: green for success and red for failure.
- MarkerCluster is used because many launch records share the same site coordinates.
- This map makes relative site-level success patterns easier to inspect.



Selected Launch Site Proximity Analysis

- We add MousePosition, a selected point, a PolyLine, and a distance label for proximity analysis.
- The provided example uses coastline coordinates near CCAFS LC-40.

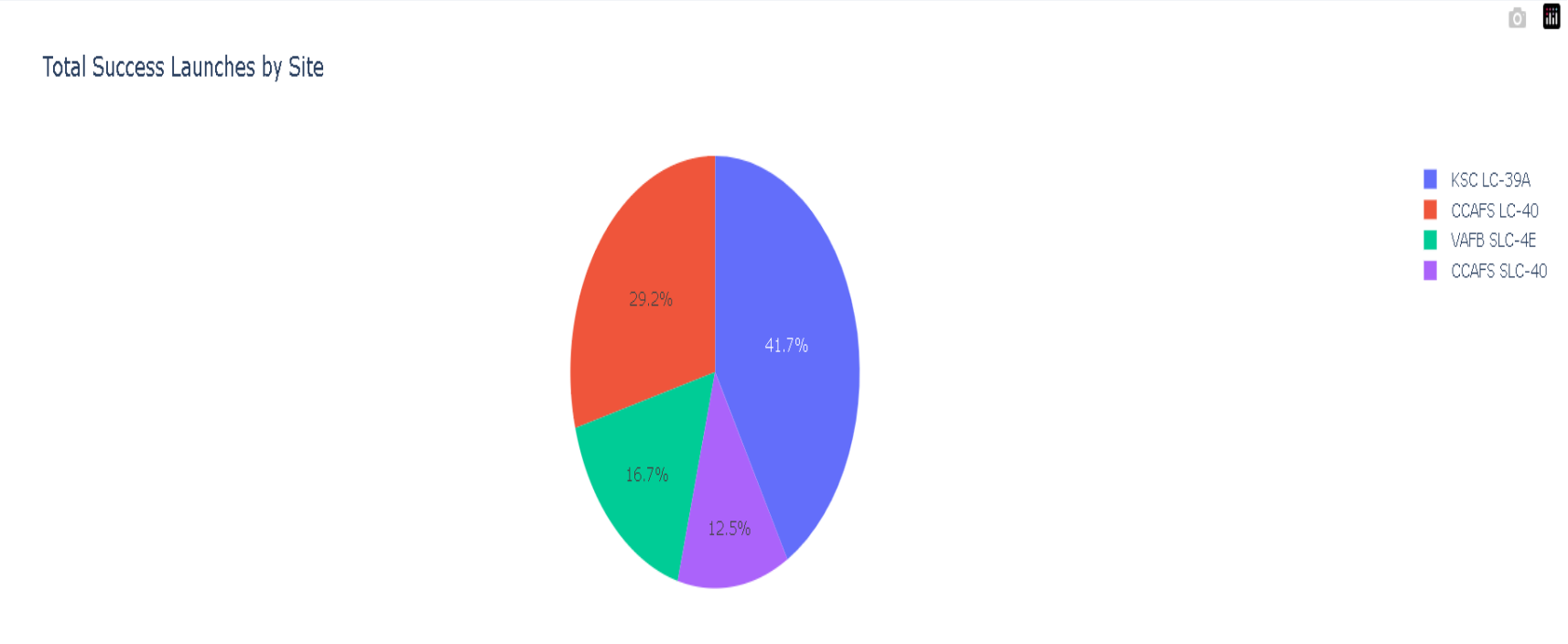


Section 4

Build a Dashboard with Plotly Dash

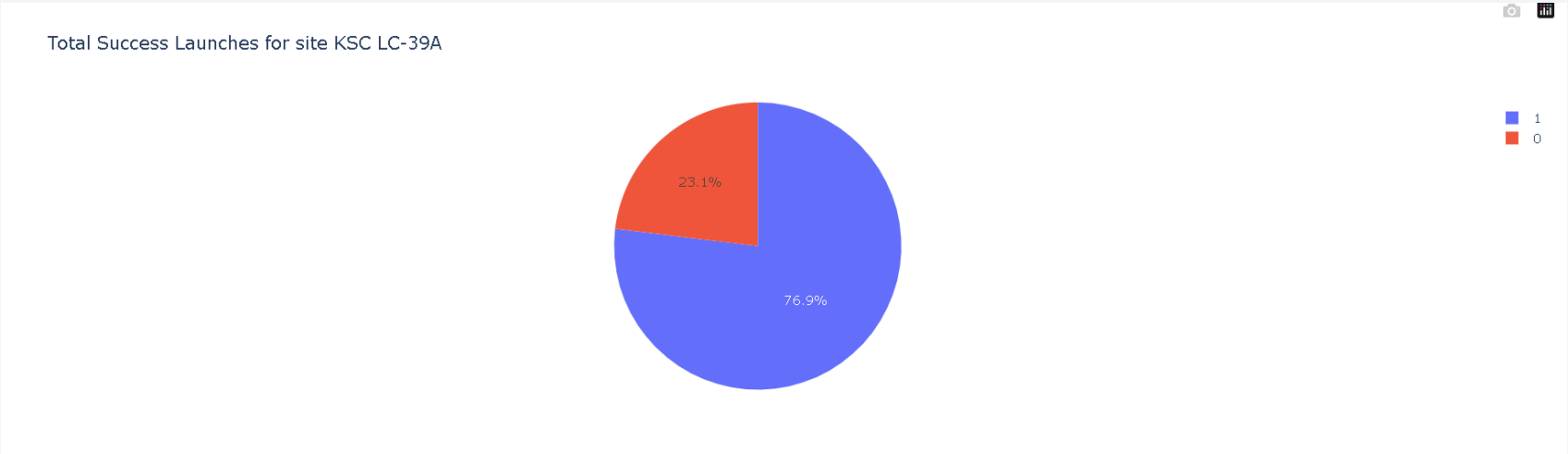
Launch Success Count Across All Sites

- The dashboard includes a launch-site dropdown with default value set to ALL.
- When entered_site == 'ALL', the pie chart shows Total Success Launches by Site.
- This view is used to compare successful launches across all sites in one chart.



Success vs. Failure at a Selected Launch Site

- When a specific site is selected, the dataset is filtered to that launch site only.
- The pie chart then shows grouped counts of the class variable.
- This interaction is used to compare success vs failure for a single launch site.



Payload Mass vs. Launch Success

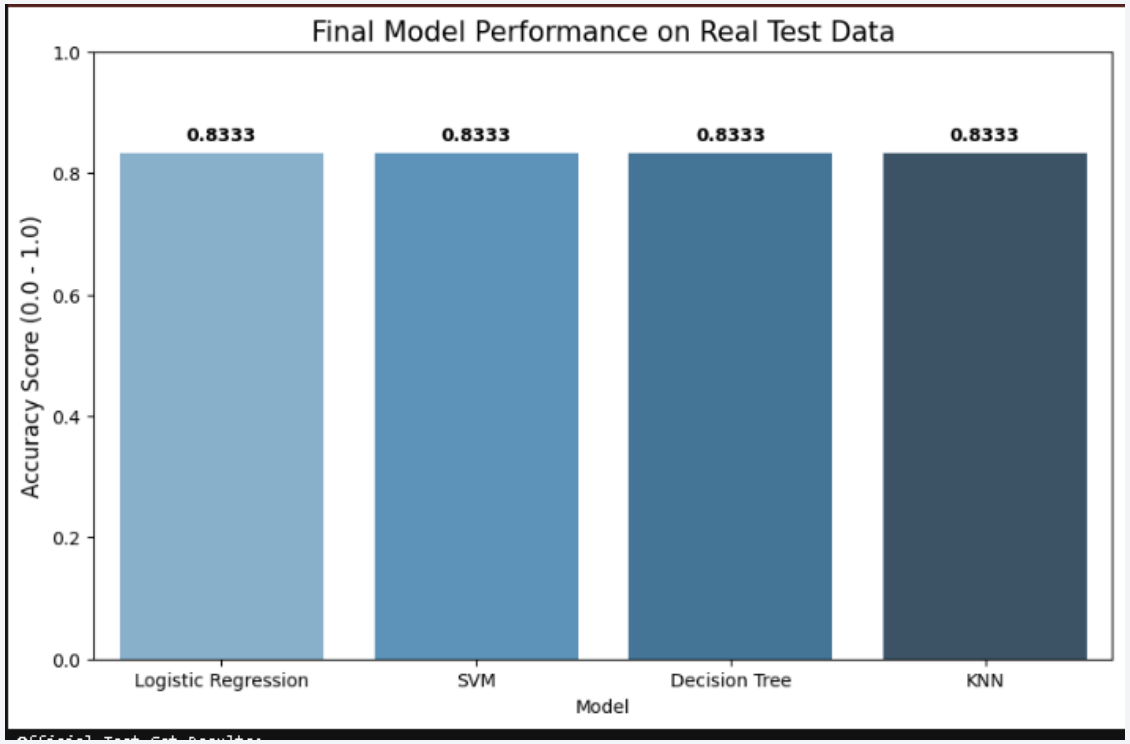
- The dashboard includes a RangeSlider from 0 to 10000 kg.
- The scatter plot maps Payload Mass (kg) on the x-axis and class on the y-axis.
- Points are colored by Booster Version Category.
- The plot updates for all sites or a selected site, depending on the dropdown and slider settings.

Section 5

Predictive Analysis (Classification)

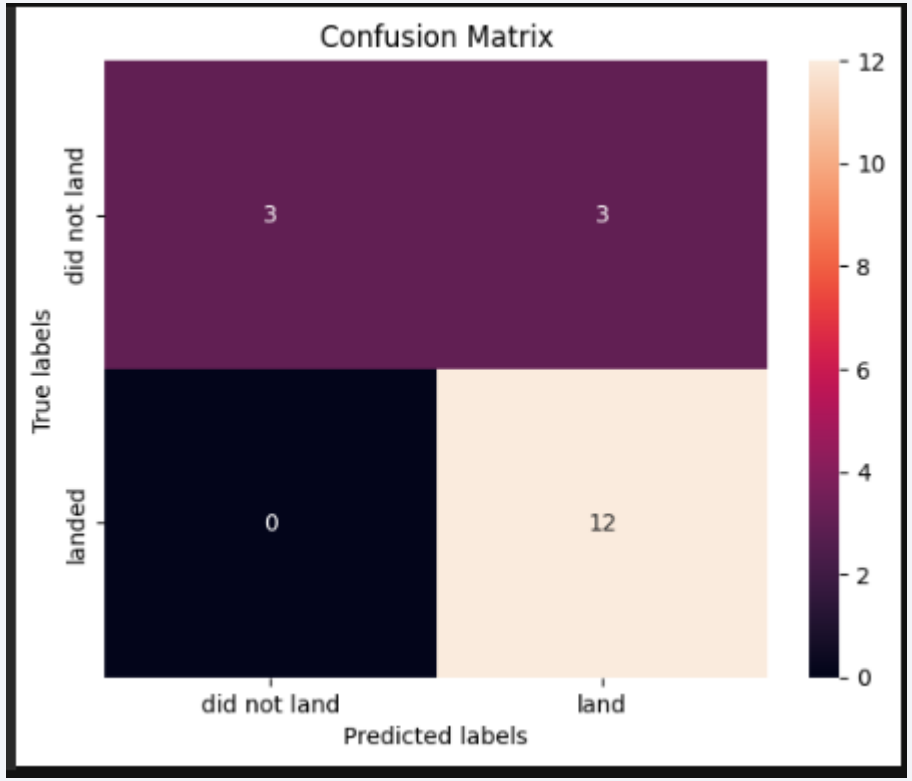
Classification Accuracy

- Logistic Regression test accuracy: 0.8333
- SVM test accuracy: 0.8333
- Decision Tree test accuracy: 0.8333
- KNN test accuracy: 0.8333
- Highest test accuracy is a tie between Logistic Regression, SVM, Decision tree and KNN.



Confusion Matrix

- Logistic Regression is used here because it is tied for highest test accuracy.
- The model distinguishes the two classes reasonably well.



Conclusions

- The data was collected from SpaceX API and Wikipedia scraping, then prepared for analysis.
- EDA showed that launch success relates to variables such as flight number, payload mass, orbit type, and launch site.
- The yearly trend plot showed that launch success improved from 2013 to 2020.
- Folium and Dash added interactive geographic and dashboard-based analysis.
- In test accuracy, all models tied at 0.8333, while Decision Tree achieved the highest cross-validation score.

Thank you!